Biochemistry Cumulative Examination

December 7, 1996

Your recently discovered allosteric enzyme is highly sensitive to ionic strength. It is tetrameric in 1 mM salt but monomeric in 100 mM salt. There is no indication that the dissociation is accompanied by confomation changes.

You have just measured the binding affinity of an allosteric inhibitor (I) of the enzyme at high and low ionic strength by equlibrium dialysis. The total concentration of protein was the same in both experiments.

1. (25) Draw a plot of fraction of protein sites occupied, Y, *vs.* [I] at high and low salt concentration. Label significant points on the plot i.e. plateau areas, inflection points, midpoints etc. Identify the physical significance of these points.

2. (25) Write equilibria expressions that describe the binding of I to the protein under both conditions. What assumptions if any did you make?

3. (25) How do these equilibria expressions account for the shape of the binding curves? At low [I] the apparent affinity of the tetrameric enzyme is lower than the apparent affinity of the monomeric enzyme. Explain.

4. (25) When you made Hill plots (log Y/(1-Y) *vs.* log [I]) of the data between 10-90% saturation, you found that plots at both high and low ionic strength were essentially linear. However, the slopes were different. What value or range of values would you expect for the two plots. Explain.